

Raw Weather → Cleaned Weather Pipeline

1. xarray Crash Course (on Dev's real ERA5 file, era5_merged.nc)

File structure confirmed:

- Dimensions: time (13,104 hourly steps), latitude (3 points), longitude (3 points)
- Variables: u10, v10 (wind), d2m (dewpoint), t2m (temperature) — all in Kelvin/m·s⁻¹
- Radiation variables (ssrd, ssr, strd, str, fdir) also present, in J·m⁻²

Conversion to pandas: confirmed working —

`ds[...].to_dataframe().reset_index()` produces a clean 117,936-row table (13,104 timesteps × 9 grid points).

2. Cleaning Pipeline (raw weather → cleaned weather)

Step	Formula	Applied to
Temperature	$^{\circ}\text{C} = \text{K} - 273.15$	t2m, d2m
Wind speed	$\sqrt{u10^2 + v10^2}$	u10, v10
Relative Humidity	Magnus formula using temp + dewpoint	temp_C, dewpoint_C

Output: cleaned_weather.csv — 117,936 rows, columns: time, latitude, longitude, temp_C, dewpoint_C, wind_speed, RH.

3. Sanity Checks (all passed)

Check	Result
Temperature range	19.53°C to 45.22°C — sensible for Ahmedabad Apr–Jun
Dewpoint range	1.73°C to 28.97°C — sensible

Wind speed range	0.013 to 13.05 m/s — sensible, no negatives
RH range	10.99% to 99.91% — within valid 0–100% bounds
Negative wind speeds	0
RH out of range	0
Missing values (any column)	0

4. Cross-Validation Against Dev's Pipeline

Merged Shweta's independently-calculated RH, wind speed, and temperature against Dev's own `utci_hourly.csv` output on matching time/location.

Result: 0.0 mean absolute difference across all three variables.

This means Dev's weather-cleaning implementation (inside his UTCI pipeline) and Shweta's independent implementation agree exactly — a strong, real cross-check confirming there's no silent calculation bug in the core pipeline.

Handoff

- `cleaned_weather.csv` — reference/backup cleaned dataset, available for anyone on the team who needs raw cleaned weather without going through the full UTCI pipeline
- This validation result goes in the Decision Log as evidence of pipeline correctness — useful if a judge asks "how do you know your calculations are right?"